Backtest of ICES Advice for Selected Northeast Atlantic Stocks
Case-study Collation and Screening

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July 21, 2026

Summary

Six candidate Northeast Atlantic case-study stocks were proposed for a *backtest* of ICES advice: a retrospective Management Strategy Evaluation (MSE) asking what would have happened if scientific advice had been followed perfectly over the historical period.

Data were collated from ICES Stock Assessment Graphs (SAG) and the advice database (ASD), together with their age-based assessments. Status, reference-point stability, and catch versus advice were reviewed for each stock, with open-loop projections at the advised target fishing mortality (no HCR feedback).

Main findings.

- **Advice–implementation gaps** for several stocks, notably Celtic and Irish Sea cod and whiting (advised catch reductions not fully realised) and mackerel (catch consistently above advice).
- **Stock status** is poor for most case studies: most stocks are below B_{trigger}; Celtic Sea cod, haddock, and mackerel show $F > F_{\text{MSY}}$ in the latest assessment years.
- **Assessment revision** matters: SSB perception and reference points shift across assessment years, especially at benchmarks.
- **Recruitment** strongly affects trajectories; screening projections at fixed target F are sensitive to mean recruitment versus recruitment deviates from a stock–recruitment relationship.
- **Northeast Atlantic mackerel** needs further work: the recent benchmark shifted scale and reference points; interpret status, advice, and projections provisionally until benchmark handling is agreed.

Decision. Proceed with all six stocks; mackerel retained provisionally pending benchmark-year handling. No alternative list is proposed.

Priority for conditioning. Operating-model conditioning depends critically on the *stock–recruitment relationship.* Biology and selectivity can follow the assessment; the key choice is how recruitment and deviates from the fitted relationship are treated in projection.

Next steps. Agree peel windows; code contemporaneous HCRs with year-specific reference points; reconstruct annual assessments; run perfect-management counterfactuals; revisit mackerel. Short-term reference projections from the latest assessment (project plan) remain outstanding and are distinct from the whole-period screening projections in this note.

1 Introduction

Evaluation of scientific management frameworks requires answering two distinct questions namely whether: (i) the assessment and advice rule are robust when applied as intended, and (ii) do realised outcomes differ from expectation because advice was not followed. A retrospective Management Strategy Evaluation (MSE) — i.e. a *backtest* of the advice rule over the historical period — can address both questions by simulating compliance with the scientific advice.

A first step is screening the case-study stocks, to evaluate whether there is an advice–implementation gap, and whether stock status is sufficient that following advice might have improved outcomes. If advice had always been followed perfectly, a backtest would only test scientific robustness, not management effectiveness. In addition, projections at a constant target F (open-loop, without feedback) were conducted to provide an additional check of whether fishing or environmental variability determine stock trends.

The workflow for screening of candidate stocks is:

1. **Collate** candidate age-structured stock assessments (Category 1) and advice from ICES.
2. **Summarise** stock status and trends (SSB and F relative to B_{trigger} and F_{target}), recruitment, and any catch–advice mismatch.
3. **Decide** whether to continue with the initially agreed stocks, or to consider alternatives.

Terminology used throughout is defined in Table 5 Appendix C.

2 Methods

A backtest is an MSE (a closed-loop simulation with feedback control) applied retrospectively over a historical window. It requires stocks with documented assessments and advice, and a period when scientific advice was not always implemented so that *following advice* becomes a counterfactual hypothesis.

The operating model representing *true* stock dynamics is taken from the most recent assessment. A retrospective peel is then run for each historical year t:

1. **Management procedure:** reconstruct stock status using only data and reference points available at t (the contemporaneous assessment).
2. **Advice:** apply the harvest control rule in force at t, with that year's reference points, to obtain catch advice for t+1.
3. **Counterfactual:** project the stock forward under perfect implementation (realised catch equal to advised catch; no TAC overshoot, override, or implementation error).
4. **Evaluation:** contrast SSB, F, and catch with the realised history over the same years.

Repeating this for successive peels t by stepping back in time builds a counterfactual trajectory under consistent adherence to advice as provided at the time.

2.1 Case studies and data

Six candidate ICES Category 1 stocks with age-based assessments were collated (Table 1). Advice follows the ICES MSY approach, except for Northeast Atlantic mackerel, where the

Coastal States MSY rule applies. Between seven and ten past assessments per stock are available in the SAG archive. Irish Sea whiting has the sparsest recent SAG coverage among the six since recent advice is zero catch meaning no assessment can be run.

Table 1: SAG assessment-year coverage for candidate stocks.

Stock	SID	Assessment years
Irish Sea cod	cod.27.7a	2017-2026
Irish Sea whiting	whg.27.7a	2016-2025
Celtic Sea cod	cod.27.7e-k	2017-2026
Celtic Sea whiting	whg.27.7b-ce-k	2017-2026
Celtic Sea haddock	had.27.7b-k	2017-2026
NE Atlantic mackerel	mac.27.nea	2017-2025

Datasets were the most recent ICES assessments and multi-year Stock Assessment Graph (SAG) downloads (assessment years range from 2016 to 2026) for stock status, reference points, and historical advice. Agreement between SAG and local `FLStock` assessments is summarised in Appendix A.

2.2 Screening projections

Open-loop projections at constant target $F = F_{\text{MSY}}$ over the historical period are used only for screening (Section 3.4). They are broader than—and distinct from—the short-term *reference projections* from the most recent assessment described in the project plan.

3 Results

3.1 Current status

Current status relative to reference points is summarised in Table 2. Most stocks are below B_{trigger} and several show $F > F_{\text{MSY}}$ in the latest assessment year. Ideally if advice had been followed all stocks will have SSBs above B_{trigger}

Table 2: Latest SAG terminal year with relative and absolute status. SSB ratios and absolute SSB: green above B_{trigger}, amber between B_{lim} and B_{trigger}, red below B_{lim}. F and F/F_{MSY}: green at or below F_{MSY}, red above.

Stock	Ass.	Year	SSB/B_{trig}	SSB/B_{lim}	F/F_{MSY}	SSB	F	F_{MSY}	B_{lim}	B_{trig}
Irish Sea cod	2026	2026	0.28	0.38	0.05	3,584	0.009	0.171	9,364	13,012
Irish Sea whiting	2025	2024	0.53	0.73	4.14	1,223	0.870	0.210	1,670	2,322
Celtic Sea cod	2026	2025	0.02	0.02	4.95	99	1.435	0.290	4,200	5,800
Celtic Sea whiting	2026	2025	0.26	0.36	0.65	13,014	0.242	0.375	36,571	50,818
Celtic Sea haddock	2026	2025	0.63	0.88	1.78	8,099	0.627	0.353	9,227	12,822
NE Atlantic mackerel	2025	2024	0.77	1.03	1.39	3,165,538	0.266	0.191	3,067,017	4,119,337

3.2 Stock assessments

ICES conducts *benchmark* assessments, in which methods and assumptions are revised, and *updates*, in which additional years of data are added to the latest benchmark.

Contemporaneous SSB estimates for each assessment year (Figure 1) show the familiar revision fan as successive assessments re-estimate the stock.

Figure 1: Spawning stock biomass by assessment year for each case-study stock.

Figure 2 plots the reference points by year (F_{MSY}, B_{lim}, B_{trigger}) relative to the latest assessment; values near 1 indicate little revision between assessment years.

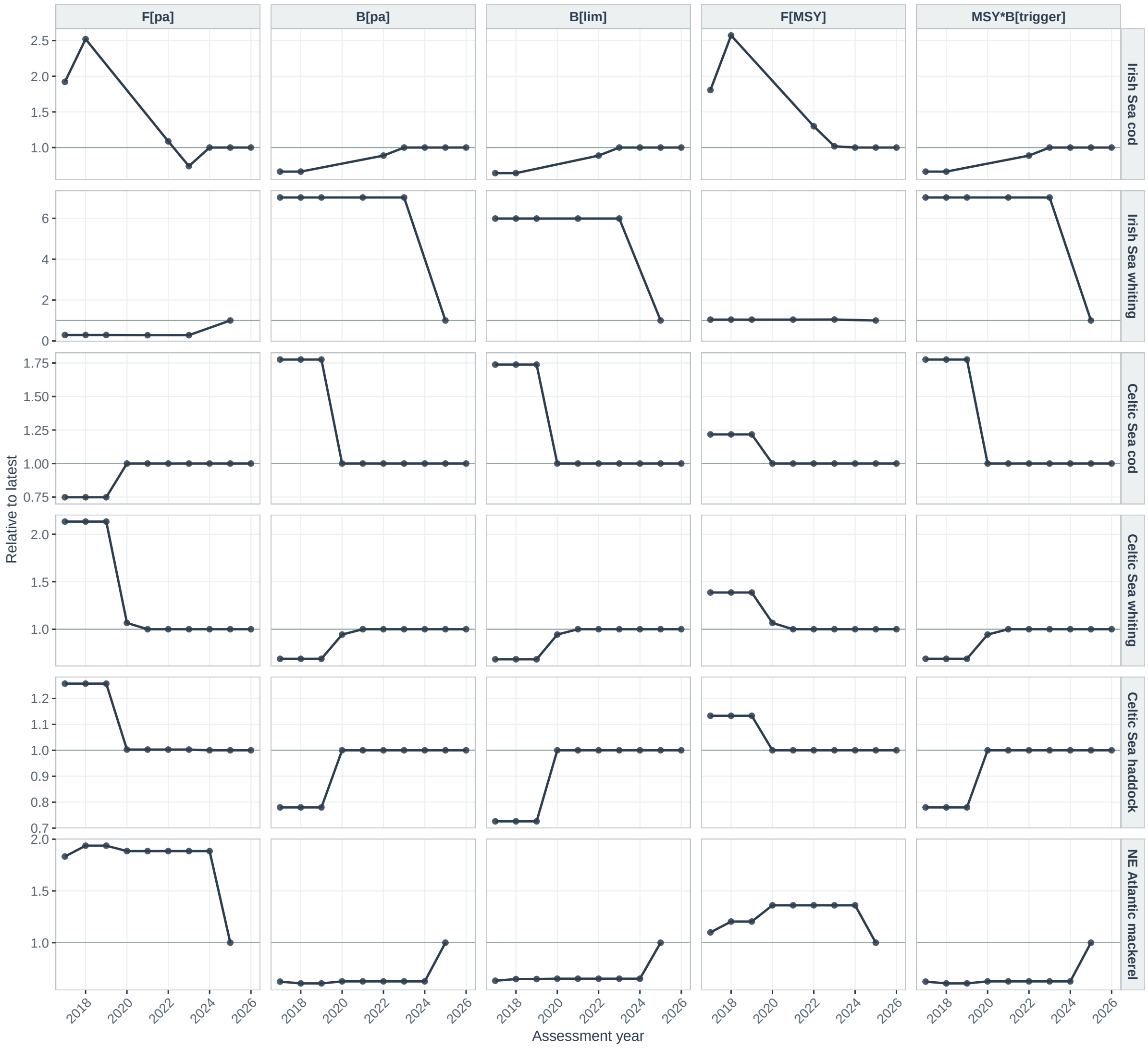

Figure 2: Reference points relative to the latest assessment year (values near 1 indicate little revision).

For several stocks the long-term SSB trend is similar across assessment years, with the main differences found in the terminal years — the usual retrospective pattern that affects precision at an update. Benchmarks have larger effects for mackerel, Celtic Sea whiting, and Irish Sea cod, where reference points or scale shift materially between assessment years. For **Northeast Atlantic mackerel**, the benchmark step changed the perceived stock scale and reference-point basis substantially. Differences between the relative trajectories may be caused by changes in the reference points, which is relatively easy to handle in the MSE by changing reference points accordingly, however for mackerel it looks like big changes have been seen also in the stock trend. Screening for mackerel should therefore be treated as provisional, and benchmark handling will be revisited in the conditioning and backtest steps

3.3 Advice

Reported catch (SAG) is compared with ICES advised catch in Figure 3. Years in which reported catch exceeds advice indicate periods when management did not fully implement the recommended catch. This screening comparison uses **catch versus advised catch** only; agreed TAC is not included here. TAC may be added later if needed for the full backtest; perfect-management counterfactuals already assume catch equal to advised catch with no TAC overshoot.

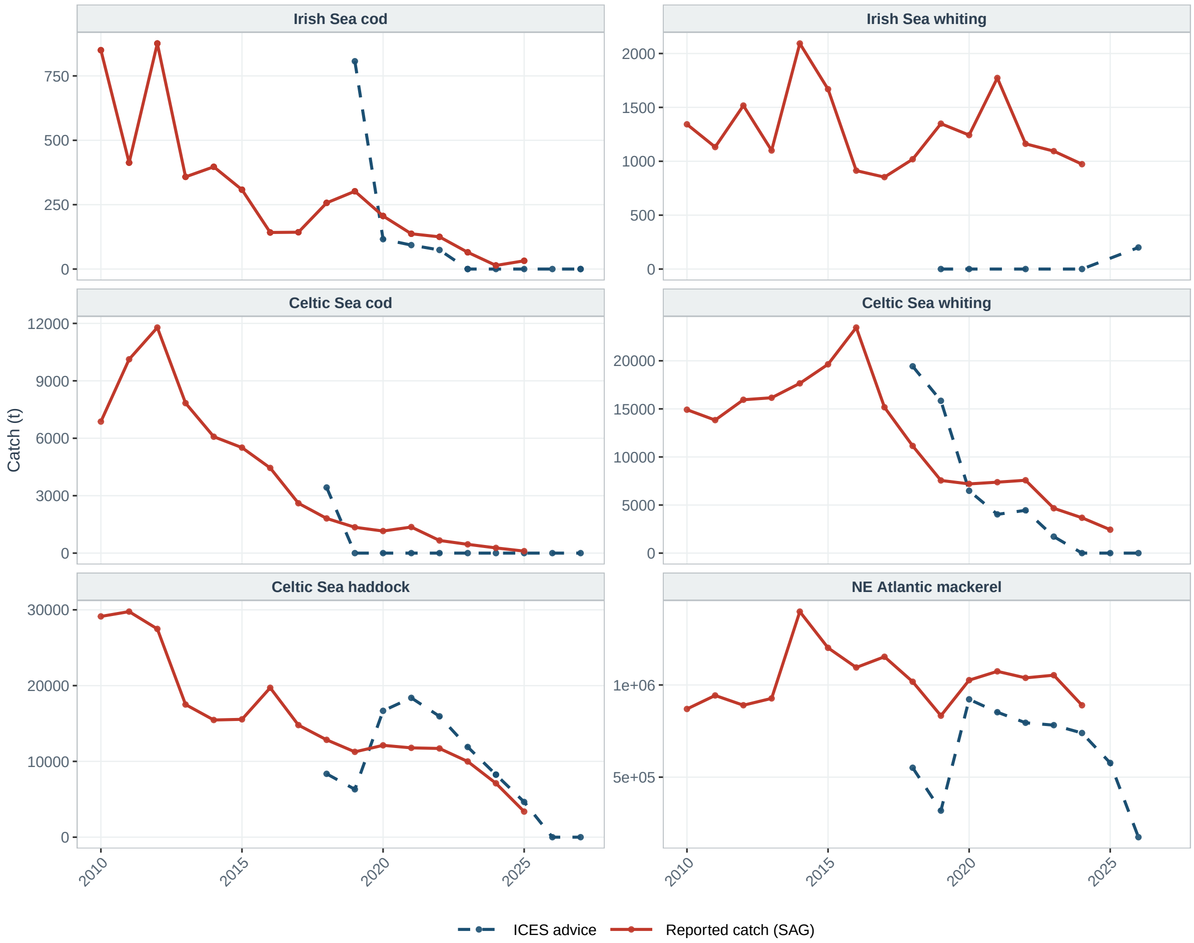

Figure 3: Reported catch (SAG) versus ICES advised catch (ASD) by calendar year of application. Facets use independent y-axes (tonnes).

For Celtic and Irish Sea cod and whiting, advice has called for large catch reductions that were not implemented. Mackerel catches have consistently exceeded advice. Celtic Sea haddock catches have generally been below the advised level.

3.4 Historical projections

Projections (open-loop) were run over the historical period at constant target $F = F_{\mathrm{MSY}}$ (Figure 4); these are used only for screening, not for interpretation. The panels show fishing mortality, biomass, recruitment, then catch. Three trajectories are shown for each stock; series are scaled by reference points or mean recruitment (horizontal line at 1). Catch is scaled by a simple MSY proxy, $MSYB_{\mathrm{trigger}} \times (1 - \exp(-F_{\mathrm{MSY}}))$.

- **Actual** (solid): realised F and recruitment from the assessment.
- **Target F, mean recruitment** (dashed): constant F_{MSY} with recruitment fixed at the geometric mean (deterministic; no year-class deviates).
- **Target F, recruitment deviates** (long dash): the same constant F_{MSY} projection with

deviates from the fitted stock–recruitment relationship, so projected year classes follow the assessment.

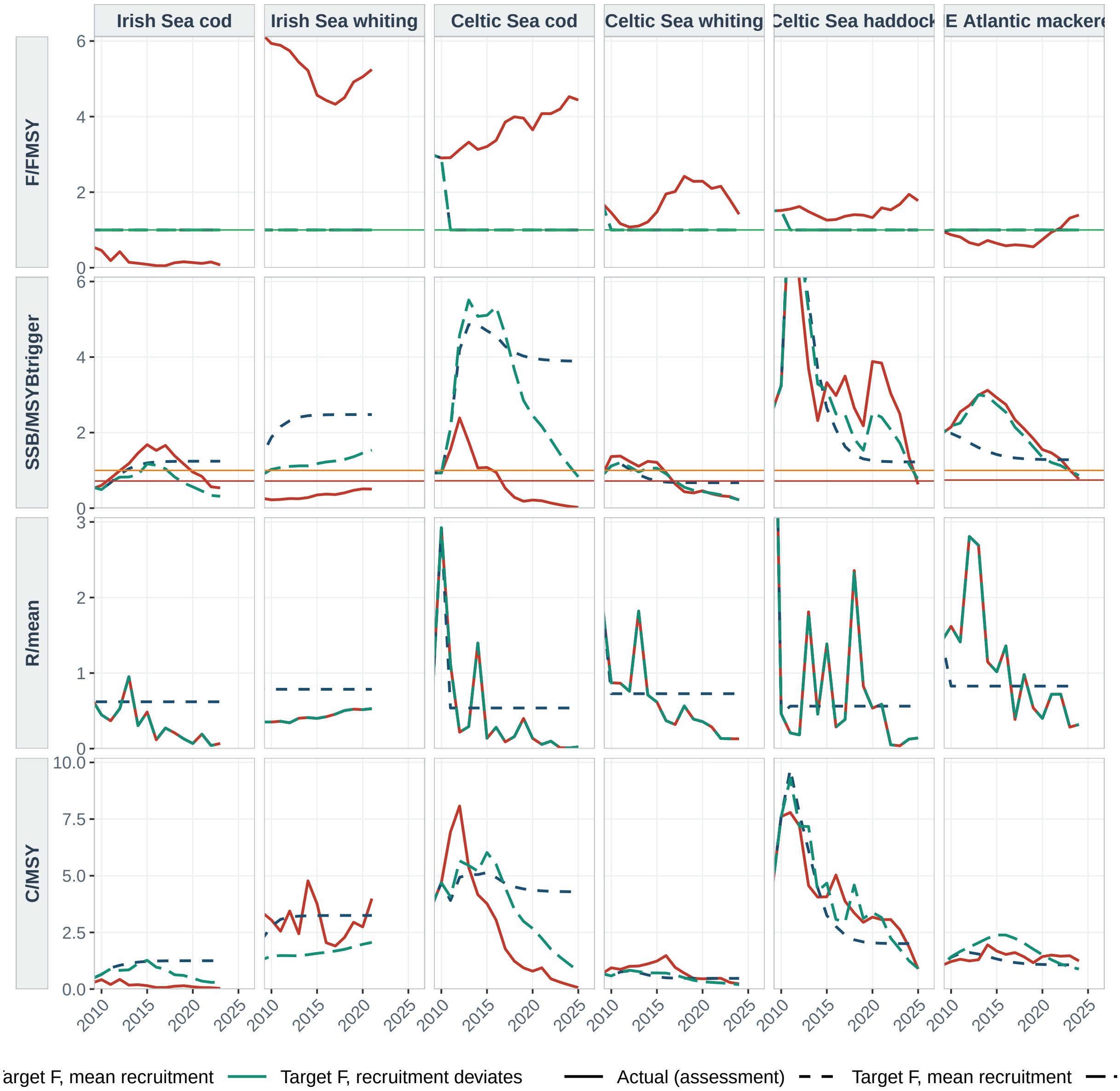

Figure 4: Actual vs. constant-target-F projections, scaled by SAG reference points. Three trajectories: Actual (solid); target F, mean recruitment (dashed; F_{MSY} with geometric-mean R); target F, recruitment deviates (long dash; same F_{MSY} with assessment deviates from the stock–recruitment relationship). Compare dashed vs. long dash to isolate recruitment; compare long dash vs. Actual to isolate realised F. Rows share a common y-axis within each metric (F/F_{MSY}: 0–6; SSB$/B_{\mathrm{trigger}}$: 0–6; $R/$mean: 0–3; $C/$MSY: 0–10). Reference lines: F_{target} (green), B_{trigger} (amber, SSB row at 1), B_{lim} (red).

Separating fishing mortality and recruitment. The dashed and long-dash projections share the same target F, and so isolate the effect of recruitment: any divergence in SSB, R, or catch between mean recruitment and recruitment deviates is attributable to year-class strength, not to F. Comparing recruitment deviates to Actual uses the same recruitment pattern but different F; divergence therefore reflects exploitation departing from F_{MSY}. Comparing mean recruitment to Actual combines both effects. This three-way contrast is for screening only (the mean-recruitment run does not use a formal stock–recruitment relationship); the backtest will condition operating models on the fitted relationship and propagate deviates in projections.

F/F_{MSY} — exploitation level The top row asks whether realised F has exceeded the MSY target. Where the **actual** trajectory lies above 1, exploitation has exceeded F_{MSY}. Celtic Sea cod, haddock, and whiting show extended periods with $F > F_{\mathrm{MSY}}$. Irish Sea cod has generally remained below F_{MSY}; mackerel was historically below the target but is now above F_{MSY} in the latest assessment year.

SSB/B_{trigger} — biomass response The second row links fishing pressure to stock size. Stocks with sustained $F > F_{\mathrm{MSY}}$ tend to remain below B_{trigger} or B_{lim}, or to show substantial declines. Irish Sea whiting has increased recently despite rising F, suggesting that recruitment or environmental drivers may also be influencing the trajectory.

R/mean — recruitment The dashed line (mean recruitment) is flat by construction, whereas the long-dash line varies with assessment deviates. Where mean recruitment and recruitment deviates diverge strongly, year-class strength — rather than F alone — drives much of the dynamics. Where the long-dash trajectory is closer to **Actual** than the dashed track, recruitment deviates help explain the historical path; where Actual differs from the long-dash run, realised F is the main driver.

C/MSY — consequences for catch The bottom row translates status and recruitment into yield relative to an MSY proxy. Overfishing, poor recruitment, or both may depress realised catch.

Summary

- **Celtic Sea cod** — $F \gg F_{\mathrm{MSY}}$, SSB well below B_{lim}, weak recruitment; catch depressed.
- **Celtic Sea whiting** — below B_{trigger} with $F < F_{\mathrm{MSY}}$; mean-recruitment and recruitment-deviate runs diverge strongly.
- **Celtic Sea haddock** and **mackerel** — below B_{trigger} with $F > F_{\mathrm{MSY}}$ (mackerel: interpret with benchmark caveat above).
- **Irish Sea cod** — low recent F but biomass still poor; terminal SSB driven down by weak year classes since 2020.
- **Irish Sea whiting** — sparse recent SAG coverage; retained for the backtest with diagnostics flagged in the reconstruction step.

Is continuing with these six stocks worthwhile i.e. is there a clear advice–implementation gap (Figure 3), and is there evidence that following advice could improve status and yield? The constant-F_{MSY} projections provide an initial quantitative check. The next stage is to run the closed-loop backtest (contemporaneous peels and perfect-management HCR counterfactuals):

- Where realised F has exceeded F_{MSY} and reported catch has exceeded advice, constant-F_{MSY} paths typically suggest higher SSB and, over medium horizons, more stable or higher cumulative yield than the depleted history—Celtic Sea cod is the clearest screening example.

- Where recent advice has been for very low or zero catch, the counterfactual is deliberately conservative on yield in the short term; the screening case for retention is then poor status and a large advice–catch gap, not short-term catch gains.
- Stocks with mixed status (below B_{trigger} but $F < F_{\text{MSY}}$, or recovering biomass) remain useful contrasts for the cross-stock synthesis.

The contrasts support keeping all six stocks for a cross-stock synthesis of non-adherence to advice. **Northeast Atlantic mackerel** is retained only provisionally: the recent benchmark break means its status ratios, catch–advice comparison, and projections must be revisited before conditioning. Screening projections used a geometric-mean recruitment proxy rather than a formal stock–recruitment relationship. For the backtest, operating models will be conditioned on assessment-consistent stock–recruitment relationships, with recruitment deviates used to represent year-class strength in counterfactual projections. **Proceed with all six initial stocks**; no alternative list is proposed at this stage.

4 Discussion

The screening results illustrate four features that the backtest should separate when attributing outcomes to advice adherence versus assessment and reference-point revision.

Perception changes at each update or benchmark. SAG SSB fans (Figure 1) show that the historical perception of stock status is revised whenever the assessment is updated, and more substantially at benchmarks. The same calendar year can therefore be judged as above or below B_{trigger} (or B_{lim}) depending on which assessment year is used. A perfect-management counterfactual that applies advice based on the contemporaneous perception must therefore be anchored to the assessment available *at the time*, not to the latest revised history. Comparing realised catches with advice derived from today's assessment could conflate implementation failure with retrospective revision of status.

Recruitment and the stock–recruitment relationship. Historical projections at fixed target F (Figure 4) show that deterministic mean recruitment often fails to track assessed SSB and catch, whereas runs that retain recruitment deviates from the stock–recruitment relationship stay closer to the assessment on recruitment-driven quantities. For several stocks, year-class strength dominates short-term trajectories relative to the choice of target F alone. Conditioning the operating model will therefore centre on the stock–recruitment relationship: its functional form, estimated parameters, and the recruitment deviates applied in projection. The backtest should report sensitivity to these assumptions so that differences between realised and counterfactual paths are not attributed solely to whether advice was followed.

Mackerel and benchmark breaks. Northeast Atlantic mackerel stands out among the case studies because the reference-point and scale revision at the recent benchmark is large relative to other stocks. Status ratios, catch–advice comparisons, and screening projections for mackerel should be interpreted cautiously until benchmark-year handling (reference points, perception, and advice rule) is agreed for the backtest peel sequence. This stock will be revisited early in the conditioning phase.

Reference points change at benchmarks. Relative reference-point series (Figure 2) confirm that F_{MSY}, B_{lim}, and $MSY\,B_{\text{trigger}}$ are not fixed for the duration of a management history:

they jump when benchmarks redefine biology, selectivity, or the basis for MSY. Advice rules coded with today's reference points could misrepresent the harvest control rule in force in earlier years. The backtest should carry forward the reference points published with each assessment year (or the values used in that year's advice) when applying the MSY rule, and treat benchmark years as structural breaks when interpreting changes in advised F or catch.

5 Next steps

The following steps follow on from the screening. Reconstruct annual perceptions and advice rules as they were, project under perfect implementation, and compare with realised history. This separates assessment revision, recruitment stochasticity, and reference-point change from questions of advice adherence. Operating-model conditioning will prioritise the stock–recruitment relationship and recruitment deviates for each stock (Section 4).

Table 3: Tass to conduct backtest.

Topic	Conclusion
Case studies	Keep all six proposal stocks; mackerel provisional pending benchmark handling
Advice rules	ICES MSY approach for gadoids; Coastal States / ICES MSY rule for mackerel
Catch vs advice	Screening uses catch vs advised catch; TAC not considered in Figure 3
Backtest window	Align peels with each stock's SAG coverage, i.e. go back to 2016
Stock–recruitment relationship	Key conditioning step: fit assessment-consistent relationship; use recruitment deviates in projections
Advice Rule	Short-term reference projections for contemporaneous assessment using reference points available at the time

1. **Agree the backtest peel window per stock.** Set start and end years based on Table 1. SAG online coverage is the binding constraint: assessment histories span only about 7–10 years (roughly 2016–2026) for most stocks, even though the local `FLStock` series run back to the 1960s–1980s. The effective backtest window is therefore about ten assessment years, which is adequate for the aims of this study and bounds how much can be concluded from the retrospective.

2. **Treat benchmarks as structural breaks.** Flag benchmark years for reference-point and model-setting changes; prioritise mackerel (Section 4).

3. **Code contemporaneous advice rules.** Implement the ICES Category 1 MSY rule for gadoids and the Coastal States / ICES MSY rule for mackerel, using each year's published F_{MSY}, $MSY B_{\text{trigger}}$, and B_{lim} (Section 4), not a single latest set of reference points.

4. **Assessment reconstruction.** Approximate annual perceptions by re-running SAM off the available inputs (most recent assessment and 2020), simulating survey CPUE where contemporaneous indices are missing. Because this is a simulation study rather than a fresh statutory assessment, the held catch-at-age and biological data are sufficient; genuine year-by-year re-runs are not required (see *Feasibility and data availability* below).

5. **Condition operating models.** Fit the stock–recruitment relationship for each stock (dynamic-B_0 or assessment-equivalent form), validate recruitment deviates, and document

sensitivity scenarios.

6. **Perfect-management counterfactuals.** From each peel, apply that year's advice and project forward under catches equal to advised catch. Report SSB, F, and catch paths against realised history, using the conditioned stock–recruitment relationship and appropriate recruitment deviates.

7. **Stock-specific summaries and cross-stock synthesis.** Tabulate time below B_{lim} and B_{trigger}, F relative to F_{MSY}, and yield/stability metrics for realised vs. counterfactual paths; synthesise where adherence would have mattered most versus where assessment or reference-point revision dominate.

Feasibility The biggest task, assessment reconstruction, depends on creating the observation error model, i.e. simulation of the survey indices, catch-at-age, and biological data as they stood at each historical assessment. SAG provides only the estimated *output* time series, not these inputs. This is a simulation exercise, not a statutory re-assessment, so contemporaneous inputs need not be recovered exactly. Selectivity-at-age and biological data can be taken from the most recent assessment, and survey indices can be simulated and SAM re-run to produce plausible contemporaneous perceptions. Two stock-specific issues remain: Irish Sea whiting has some missing SAG years (an easy fix), and Northeast Atlantic mackerel has a large benchmark change (needs more thought).

Yimeline. These next steps map onto the project plan's Tasks 2–4—assessment reconstruction (20 days), counterfactual simulation (15 days), and comparative analysis and reporting (10 days)—about 45 days in total, following the completed Task 1 screening (5 days). However, much more work was conducting during the screening that first envisaged; much of the code (SAG/`FLStock` for processing the ICES datasets, conditioning the operating model, projections, HCR backtests, and plotting is in place; remaining effort is for running the code, checking results, and drafting the report, so the plan remains on track.

Immediate actions.

- Confirm the stock list: all six candidates retained (mackerel provisional).
- Start assessment reconstruction for one prototype stock.
- Produce short-term reference projections from the most recent assessment (project-plan Task 1 remainder), distinct from the whole-period screening projections in Section 3.4.
- Revisit Northeast Atlantic mackerel: agree benchmark-year treatment and reference-point carry-forward before conditioning the operating model.

A Appendix: SAG versus local `FLStock`

Scaled trajectories from the latest SAG assessment year compared with the local WGCSE `FLStock` objects (common SAG reference points: SSB/B_{trigger}, F/F_{MSY}, recruitment relative to mean SAG recruitment, and catch relative to a simple MSY proxy).

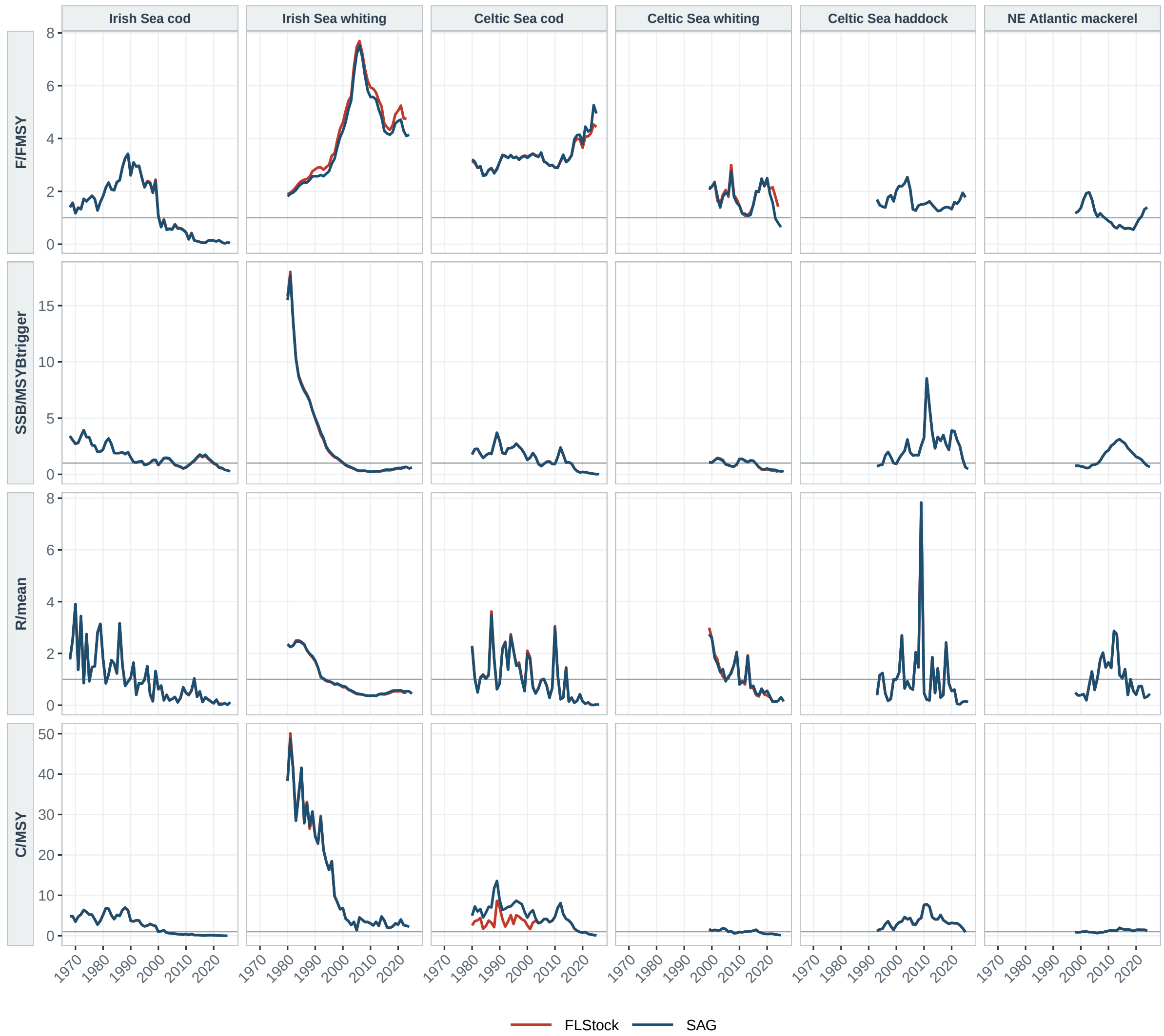

Figure 5: SAG vs. local `FLStock`, scaled by SAG reference points (horizontal line at 1).

B Appendix: Software and reproducibility

Analysis uses FLR (`FLCore`, `FLBRP`, `FLasher`, `ggplotFL`) and the `FLBacktest` / `backtest` tools (`hcrICES`, projection and peel helpers). The notebooks write `RData`; this note and later reports load those results.

Table 4: Paths used for the screening deliverable.

Path	Role
`Rmd/01_screening.Rmd`	Screening, SAG vs FLStock, target-F projections → `screening.RData`
`Rmd/02_condition_om.Rmd`	OM conditioning → `om.RData`
`Rmd/03_backtest.Rmd`	HCR counterfactuals → `backtest.RData`
`scripts/build_week1_tex.R`	Regenerates tables/figures for this document
`tex/backtest_screening.tex`	This deliverable
`data/reference/stocks.csv`	Stock metadata and advice years

C Appendix: Terminology

Table 5: Terminology used in this document.

Term	Meaning
SSB	Spawning stock biomass
F	Fishing mortality (mean F across ages)
F_{MSY}	MSY-related fishing mortality (from SAG when available)
F_{target}	Target fishing mortality used in advice (from SAG when available)
B_{lim}	Limit biomass reference point
B_{trigger} ($MSY B_{\text{trigger}}$)	Trigger biomass reference point
SAG	ICES Stock Assessment Graphs
ASD	ICES advice database
HCR	Harvest control rule
OM	Operating model used in MSE
Stock–recruitment relationship	How recruitment depends on SSB (the informal code abbreviation “SRR” is avoided in this text)
TAC	Total allowable catch (scoped out of the catch–advice comparison in Section 3.3)